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Nerve and muscle cells control their electrical activity by opening and closing molecular-sized pores in the cellular membrane called ion channels. Hence, nervous system and neuromuscular disorders may be detected and characterized by measuring ion channel function. Such is the case of psychiatric disorders, anesthesia, epilepsy, hypertension, and heart arrhythmia, among others. Conversely, these disorders may be treated by drugs that bind to ion channels. The ability to measure ion channel function in outer space is thus very important for the purpose of future long-term space exploration requiring in-flight sophisticated experimentation or therapeutic screening. Unfortunately, measuring ion channel function is a time-consuming, vibration-sensitive task involving bulky, heavy equipment which is impractical or virtually impossible to implement for outer space-based experimentation. Ion channel function is most often measured by means of in-vitro electrophysiological recordings that must be obtained from single cells or cell membrane fragments by an extremely low throughput process based on micropositioning a pipette tip into intimate contact with the cell membrane. Here we propose an alternative approach consisting of immobilizing cells and cellular

Task Description:	membrane patches in an addressable microarray. The array will contain microfluidic channels buried underneath the cells acting as "embedded pipettes" that will establish electrical contact with the cell through a nanofabricated aperture. This approach will allow for rapid electrophysiological screening of large number of putative drugs and for highly parallel diagnostics of ion channel gating kinetics. The measurement of ion channel activity is a tedious, costly task. Briefly, it involves obtaining electrophysiological recordings from single cells or cell membrane fragments by means of a sharp glass pipette (~1 μm tip aperture diam.) carefully sealed against the cell membrane. The goal of this project is to build a microdevice that will automate the sealing process on large numbers of cells simultaneously, thus allowing for miniaturized, high-throughput electrophysiological recordings. We propose a radically different approach to electrophysiological recordings based on a microfluidic circuitry under the cell culture substrate that accesses the cell's bottom surface through micron- or submicron-sized holes. Gentle suction applied through the microchannels will cause the cell membrane to seal against the nanohole, and the cells will be "torn" away while applying suction to the holes, effectively leaving a membrane patch covering the hole. A microelectrode integrated with each microfluidic channel will monitor ion flux through the membrane patches as they are exposed to various ion channel-targeting compounds. The device is designed to allow for preparation of the membrane patches on the ground and for fully automated operation during in-flight recording. Unlike the traditional method, this approach straightforwardly allows for multiple simultaneous experiments with minimum amounts of fluids, weight, or volume. Our approach has many advantages: 1) since all the hardware associated with pipette micropositioning (micropositioners, microscope, vibration isolation equipment) is eliminated, the total weight and volume of the experimental setup is reduced by orders of magnitude; 2) it is scalable, i.e. it allows for probing many cells/membrane patches in parallel; 3) since no pipette micropositioning is necessary, experimentation time is reduced and a skilled operator is not needed; 4) the apertures and the microchannels can be straightforwardly fabricated in parallel and integrated with microelectrodes, thus reducing fabrication costs and bypassing the laborious process of pipette fabrication and setup; 5) it has no moving parts, so it is insensitive to mechanical vibrations and, for the same reason, portable; 6) potentially, low rates of success in establishing a good seal are intrinsically compensated by the large number of cells probed; and last but not least, 7) the same microchannel used to pull suction through the aperture can be used to modulate the composition of the extracellular side of the membrane by delivering various solutions (neurotransmitters, growth factors, ions, etc.). This enterprise has a component of high risk because, as stated in the original proposal, it is still uncertain whether the cell membrane will seal against a non-glass surface the way it seals against a glass pipette. Although this proposal only contemplates the ground-based implementation of a microarray of membrane patches, the device should allow for inexpensive in-flight testing of ion channel function because it will be portable, robust (vibration-insensitive), light-weight, low-volume, automated, and highly parallel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many nervous system and neuromuscular disorders are caused by alterations of ion channel activity, hence a high-throughput tool to develop drugs that target ion channels is likely to have a significant impact on human health.
Task Progress:	Funding started in August 2001. As authorized by NASA, the project was supplemented by the Whitaker Foundation and was scheduled to involve a postdoc (NASA) and a student (Whitaker). 1) Personnel and logistics. As explained in last year’s report, a master’s student was hired for the project. His excellent results earned him a UW Nanotechnology Center fellowship, which has now expired. However, the other student working on the project under Whitaker-funded support (which has expired too), received a UW Nanotechnology Center fellowship, so there are still 3 people working in the project. We originally proposed to “drill” the nanoapertures using a focused ion beam instrument being purchased by Bruce Darling (UW Electrical Eng.). Although the instrument was paid for, it was never delivered and the company has filed for bankruptcy. However, this setback forced us to come up with alternative processes to make nanoapertures, resulting in three invention disclosures to UW Technology Transfer Office. Out of these three provisional patent application, one has already been converted to a utility application with the USPTO, which is still pending. 2) Electrophysiological recordings. The students are now experts at electrophysiological recordings of various cell types (including muscle cells, CHO and HEK cells, and neurons) in traditional setups and are currently testing different device designs with cells, as explained in detail below. 3) Specific Aim #1 (Microfabrication of the device). The aim is to fabricate a cell culture substrate that contains embedded microchannels such that the cells or cell membrane patches can be accessed from underneath through a nanoaperture. We have solved all of the challenges involved in the microfabrication of the device, which is reported in a paper, accepted for publication in the Journal of Biomedical Nanotechnology [Ed. note: paper now published and noted in Bibliography--4/2010]. The fabrication process is shown in that paper. Briefly, the silicon nitride membranes are made by a) creating a silicon nitride layer (~300 nm-thick) on a silicon wafer by low pressure chemical vapor deposition (LPCVD) in a furnace; b) photolithographically patterning the silicon nitride layer on the back of the wafer so as to expose square silicon areas; and c) etching the exposed silicon from the backside with KOH until the frontside nitride layer is reached (KOH does not attack silicon nitride). The nanoholes are made on the silicon nitride membranes by electron beam lithography. 4) Specific Aim #2 (Electrophysiological recordings of live cells). The electrical characterization of the silicon device is explained in detail in the text and schematically summarized in Fig. 2 of the paper. Essentially, we found that membranes made of silicon nitride (much more robust than silicon oxide) were also just as good for electrophysiology as glass pipettes – the gold standard tool for electrophysiology. A key finding was that this is only achieved if the supported part of the silicon nitride layer (i.e. the one that rests on silicon) is isolated from electrolyte exposure. Importantly, the device is fully biocompatible with cells in culture, as demonstrated by the normal viability and morphology of cells cultured atop the membranes. The device has many possible uses as described in the manuscript. The publication of this manuscript essentially concludes Specific Aim #1 satisfactorily. Unfortunately, despite the great electrical performance of the device overall, all our attempts to produce a gigaohm seal between a cell and a nanohole in a silicon nitride membrane have been unsuccessful. We know that our expertise in electrophysiology is not questionable, since we routinely achieve gigaohm seals with pipettes as the control experiment. Given this lack of success, in the last two years we have decided to pursue a different design, entirely based on poly(dimethyl siloxane) (PDMS).

	The PDMS device fabrication process. First, electron beam lithography with SU-8 negative photoresist is used to produce alignment marks and lines with nanoscale cross-sections. Next, photolithography is used to create microscale channels to overlap the ends of the existing nanoscale channels. This results in a SU-8 master device with a network of raised microscale channels connected by nanoscale channels. PDMS is then used to replicate the SU-8 master using a soft replica molding technique. The surface of the PDMS replica is activated through an oxygen plasma treatment and bonded to a thin membrane of PDMS that has been activated by the same technique. This creates an enclosed channel network. The device is then “inflated” and recast with PDMS to create a planar surface at the cell/channel interface. Slicing the device produces a flat surface with a nanohole that is microfluidically accessible from underneath. Alternatively, the microchannels can be used to selectively deliver the cells to the nanoholes. The device has undergone characterization and is currently being tested with Rat Basophilc Leukemia (RBL-1) cells. This model cell line is frequently chosen because cells grow in suspension and present endogenous ion channels. We have been able to achieve seals between the cell membrane and the PDMS nanoholes on the order on 100 MOhms, about an order of magnitude smaller than the “gigaohm” seal that characterizes a good patch. While patch clamp recordings are possible with 100 MOhm-seals, the high noise precludes from detecting single-channel responses. We are presently investigating why the patch has a low resistance. 5) Specific Aim #3. In progress. We have applied for and were granted a no-cost extension because one of the students working on the project had a child and had to go on maternity leave.
Bibliography Type:	Description: (Last Updated: 04/27/2010)
Articles in Peer-reviewed Journals	Kosar TF, Chen C, Stucky NL, Folch A. "Arrays of microfluidically-accessible nanoholes." Journal of Biomedical Nanotechnology 2005 Jun;1(2):161-7. http://dx.doi.org/10.1166/jbn.2005.023 , Jun-2005